The interplay between OmpLA and membrane asymmetry

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Membrane asymmetry affects not only bilayer properties but also the behavior of embedded proteins. The activity of bacterial outer membrane protein LA (OmpLA), for instance, has been linked to differential stress across the leaflets of asymmetric membranes (1). Conversely, membrane proteins such as OmpLA are known to modulate the properties of symmetric membranes, notably through local thickness deformations (2). How these two effects interact — asymmetry shaping protein behavior, and protein presence reshaping membrane asymmetry — remains poorly understood.

To investigate this interplay, we performed atomistic MD simulations of OmpLA in four membrane compositions matching our experimental system: binary mixtures of a donor lipid (POPC or POPE) and an acceptor lipid (POPG or POPS). Each composition was simulated symmetrically and in two asymmetric configurations, corresponding to the two possible protein orientations arising during preparation of asymmetric large unilamellar vesicles (aLUVs). Comparison with protein-free counterparts allows us to assess how OmpLA alters membrane properties under symmetric versus asymmetric conditions, and how asymmetry in turn affects the local environment around the protein.

We present preliminary results from these simulations, alongside practical considerations that arose during setup — including challenges in constructing stable asymmetric bilayers and complications from using protein structures solved at cryogenic temperature. We discuss how these methodological factors may influence the interpretation of asymmetry-driven effects on membrane protein behavior.

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2. Semeraro EF, Bartoš L, Piller P, Deb R, Keller S, Vácha R, et al. Membrane Thickness Strain from Protein Inclusions: A Multiscale Simulation and X-Ray Scattering Study of Proteoliposomes [Internet]. bioRxiv; 2026 [cited 2026 Jul 29]. p. 2026.07.03.736288. Available from: https://www.biorxiv.org/content/10.64898/2026.07.03.736288v1 doi:10.64898/2026.07.03.736288